CarbonWise

What the site can do, what its results mean, and exactly how to use it.

Live site: carbon-wise-kdf7iafbypatdogdamqcsf.streamlit.app

Three ways to use the site

Choose	What you do	What the result means
Analyze my data	Upload a CSV or Excel file, map columns, review values, set a budget and priorities.	A model comparison using the values you provided. It does not verify your inputs.
Try a sample plan	Set a budget and priorities for generated example projects, then build a sample plan.	A hypothetical demonstration, not actual city proposals or measured outcomes.
Explore official evidence	Browse source-linked GCC records and public contract awards.	Shows what sources reported; it does not establish payment, completion, or causal impact.

Start here

The opening screen offers three paths. Use **Analyze my data** for your own file, **Try a sample plan** for a quick demo, or **Explore official evidence** to inspect source-backed records. These modes are deliberately separate.

Use your own project file

1. Upload a CSV, XLSX, or XLSM file. The limit is 25 MB and 500 project rows; Excel reads the first worksheet. Legacy XLS files are not supported.

2. Use one row per project and map five required fields: unique project ID, cost, annual carbon benefit, resilience benefit, and equity/community benefit. Name, category, and area are optional.

3. Enter the cost unit exactly as it appears in your file. CarbonWise does not convert units. Review the mapped table, fix errors or remove rows, then press **Use these columns**. Missing numbers are not filled in automatically.

4. Open **Compare plans**. Set a budget and priorities for carbon, resilience, and community score. Press **Generate investment plan**.

5. Review selected projects and comparison. Download mapped values, original selected rows, alternative plans, or the full run packet.

Reading your results

The output depends on which path you chose. Check the label before describing a number in a presentation.

Uploaded-data results

- Carbon and resilience benefits are sums of the input values you mapped. They are estimates supplied in your file, not measured outcomes.
- The community comparison score is scaled from 0 to 100 within the uploaded dataset. It is relative to the other rows, not an absolute measure.
- The optimizer searches for feasible portfolios within your budget and ranks trade-offs according to the priorities you set. It is decision support, not a funding decision.
- Costs must use a consistent unit. A budget entered in one unit cannot be compared safely to project costs entered in another.

Try a sample plan

Choose **Try a sample plan**, set a budget and community priorities in the sidebar, then press **Build sample plan**. The page shows the selected sample portfolio. Open **See comparisons, rationale, and fairness checks** for more detail. The optional five-budget comparison takes longer.

The 56 sample projects are generated for demonstration. They are not verified proposals, approved budgets, or measured Chennai outcomes. Some area indicators come from cited sources, but the candidate projects and resulting plans are hypothetical.

Explore official evidence

- **Overview:** counts, reported-estimate coverage, data notes, and GCC progress examples.
- **GCC project list:** filter the 23 supplied starter records, open GCC report links, and download the filtered register.
- **Contract awards:** inspect drainage and Chennai Metro Rail award records and source links.
- **Sources & checks:** expand the evidence library, calculation checks, limitations, and presenter walkthrough.

Evidence boundaries

A source-linked row means the source published that statement. Reproducing a formula checks arithmetic from stated inputs, not whether the real-world activity happened as assumed. A contract award is not proof of payment or completion. The records shown do not establish causal reductions in flooding or emissions; actual spend, completion, and measured outcomes may be unknown.

Street photos and what-if comparisons

These tools help record observations and compare scenarios. They do not turn a photo into verified project evidence.

Street photo: exact steps

1. In **Analyze my data**, open **Street photo**. Choose either **Upload a photo** or **Use camera**.

2. Add the photo. CarbonWise estimates the share of pixels in two colour ranges: green-hued and low-saturation grey. It does not identify trees, roads, standing water, safety, or project completion. Lighting, shade, and camera settings affect the estimate.

3. Select an observation you personally confirm. Link an area only if you know it; otherwise leave it unlinked. Add an optional note, tick the confirmation box, then choose **Save observation to this session**.

4. The note is held for the current browser session and does not change project data or a plan. If you later apply a resilience adjustment, it is a percentage you entered, not a computer-vision result. Download the audit packet to keep the note and photo.

Compare two runs

Generate a plan, change the budget or priorities, and generate another plan. Open **Compare runs** to see what changed. If project values changed between runs, the comparison flags that too; it may reflect changed inputs as well as changed settings.

Review and downloads

The **Downloads** section lets you keep the original active file, saved photo notes, and latest run packet. Session state is not a permanent project database. Download anything you need to preserve.

A simple live demo (about 3 minutes)

1. Start at **Start here** and explain the three modes. 2. Choose **Try a sample plan**, set a budget, and press **Build sample plan**. 3. Point to the selected rows and budget used. 4. Open the comparison panel and show trade-offs. 5. Open **Explore official evidence** and show one source-linked GCC record or award. 6. Close by explaining that reported progress, award values, generated scores, and causal impact are different things.

Suggested description

"CarbonWise is a prototype that helps inspect public climate-project records and compare project plans using supplied data. It labels reported evidence, generated examples, and user assumptions separately."

The ZIP also includes a synthetic sample CSV for practicing the upload flow. Its values are invented and must not be presented as real project evidence.